MIL-C-60230B(MU)
12 July 1973
SUPERSEDING
MIL-C-60230A(MU)
24 May 1968

MILITARY SPECIFICATION

CONTAINER, SHIPPING, REUSABLE FOR RADAR CHRONOGRAPH, M36 ACCESSORIES

1. SCOPE

1.1 This specification covers the fabrication and testing of one type container for associated equipment for Radar Chronograph Set, M36 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel, Alkyd, Semi-Gloss

Military

MIL-A-5092	Adhesive, Rubber, Synthetic and Reclaimed Rubber Base
MIL-W-8604	Welding of Aluminum Alloys, Process For
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing Of
MIL-C-21567	Compound, Silicone, Soft Film
MIL-B-23362	Brazing of Aluminum, and Aluminum Alloys
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46842	Cushioning Material, Unicellular Polyethylene Foam (For Packaging Purpose)

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-171	Finishing of Metal and Wood Surfaces
MS16535	Rivet, Tubular Oval Head

DRAWINGS

US Army, Frankford Arsenal

A8566361	Container, Shipping, Reusable, Accessory
A10539222	Valve, Relief
A10547130	Plate, Valve Instruction
B8215734	Card Holder
B8644392	Packing, Preformed
B10539157	Extrusion Edge Female
B10539158	Extrusion Edge Male
B11727515	Acrylonitrile Butadiene Styrene, Polycarbonate
C10539216	Handle Assembly
C10539217	Clamp Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as indicated by Contracting Officer.)

3. REQUIREMENTS

3.1 General.- The container assembly shall be fabricated to provide a pressure-tight container capable of protecting equipment against environmental conditions normally encountered during storage and shipment. The container shall be fabricated and assembled in accordance with the requirements specified herein and Figures 1, 1A, 2, 3, 4, 5, 6 and 7.

3.2 Material.- Material used for forming container shells and cradles shall conform to Drawing B11727515.

3.3 Shell forming process.- The upper and lower shells shown in Figures 3 and 4 shall be thermo-formed by any proven method normally associated with the plastic industry. Finished formed parts shall be entirely free of blisters, blemishes, or thermo shock stress marks.

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3.4 Fabrication.

3.4.1 Sealing extrusions.- The upper sealing extrusion shall conform to Drawing B10539157 and the lower sealing extrusion shall conform to Drawing B10539158. The extrusions shall be formed to mate with container shells by proven techniques and controlled tooling to provide uniformity in flatness and alignment as shown on Figure 5. Suitable flexible inserts shall be used to prevent distortion of cross sectional shape of extrusions and prevent chatter or cracking. The upper and lower extrusions shall be joined by butt welding or brazing which shall be ground smooth and exhibit soundness of good practices conforming to the requirements of MIL-W-8604 or MIL-B-23362.

3.4.1.1 Finish.- The formed extrusions shall be surface treated by finish 7.3 followed by finish 21.3 of MIL-STD-171, Color X24087 of FED-STD-595. Finish shall be applied prior to assembly of extrusions to container shells.

3.4.1.2 Extrusion attachment.- The sealing extrusions shall be attached to the container with a two part silicone sealant conforming to General Electric RTV 560 with Primer SS4004 or Dow Corning RTV 589 with Primer 95-013. The primer shall be applied to the mating surfaces of the shells and extrusions. The sealant shall be applied into the extrusion well and the extrusions shall be assembled to the shells to provide structural integrity and a pressure-tight joint. The sealing extrusions shall be tool rolled and crimped around the periphery to assure positive attachment to container shells.

3.4.2 Clamp assemblies.- Ten (10) clamp assemblies conforming to Part Number 10539217-2 shall be attached to lower sealing extrusion as shown on Figure 1. Attachment shall be accomplished as shown on Figure 2. Touch up shall be accomplished with paint conforming to TT-E-529, Color X24087 of FED-STD-595.

3.4.3 Handles.- Two (2) handles conforming to Part Number 10539216-2 shall be attached to lower extrusion, as shown on Figure 1. Attachment shall be accomplished as shown on Figure 2. Touch up shall be accomplished with paint conforming to TT-E-529, Color X24087 of FED-STD-595.

3.4.4. Preformed packing.- Preformed packing shall be fabricated from rubber conforming to Part No. 8644392-1. The packing shall be formed by vulcanizing or by butt bonding. Butt bonding shall be accomplished with adhesive conforming to MIL-A-5092, Type II. Prior to inserting the packing in the female sealing extrusion, adhesive conforming to MIL-A-5092, Type II shall be uniformly applied within the well. There shall be no evidence of excess adhesive in the packing well. The packing shall be surface treated with a soft film silicone compound conforming to MIL-C-21567.

3.4.5 Relief valve.- A two way automatic pressure relief valve

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conforming to Drawing A10539222 shall be installed on the cover as shown on Figure 1.

3.4.5.1 Valve instruction plate.- An instruction plate conforming to Drawing A10547130 shall be installed adjacent to relief valve in the upper sealing extrusion as shown on Figure 1.

3.4.6 Card holders.- Two (2) card holders shall be assembled to the front side of the assembled case as shown on Figure 1. The card holders shall conform to Part Numbers 8215734-1 and 8215734-2. Each card holder shall be secured in position with seven (7) rivets conforming to MS 16535-117.

3.5 Interior arrangement.

3.5.1 Description.- The interior arrangement consists of one polyethylene cushion insert in the body and one polyethylene cushion pad in the cover. The cushions are bonded in position as shown on Figure 1A. The adhesive shall conform to MIL-A-5092, Type II.

3.5.1.1 Cushioning.- Cushions shown on Figure 1A shall be fabricated as shown on Figure 6 and 7 with polyethylene foam 2 lbs/cubic foot density, color charcoal gray or black, conforming to MIL-C-46842.

3.6 Marking.- Marking legends shown on Figure 1 shall be applied by a silk screen or stencil, with paint conforming to TT-E-529, color white, 27875 of FED-STD-595.

3.7 Performance.- The reusable container fabricated to the requirements of this specification shall maintain a positive pressure of 1 psi without loss of pressure.

3.8 Environmental.

3.8.1 Storage.- The container assembly shall be capable of maintaining dimensional and structural integrity when subjected to temperature extremes of 160°F to -70°F. "The container shall show no evidence of distortion or cracks and shall remain pressure-tight."

3.8.2 Impact resistance (ambient temperature).- The assembled container shall show no evidence of cracking, chipping or delamination when dropped from a height of 24 inches. The container shall then be tested for being pressure tight.

3.8.3 Impact resistance (low temperature).- The container assembly shall show no evidence of cracking or fracture when subjected to low temperature conditioning at -40°F immediately followed by drop test from a height of 24 inches. The container shall then be tested for being pressure tight, after container is conditioned at ambient temperatures.

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3.9 Workmanship.- Containers fabricated to the requirements of this specification shall be in accordance with proven commercial forming techniques with minimum machining or routing of shells and shall be free of defects that might affect strength, serviceability, and protection of contents.

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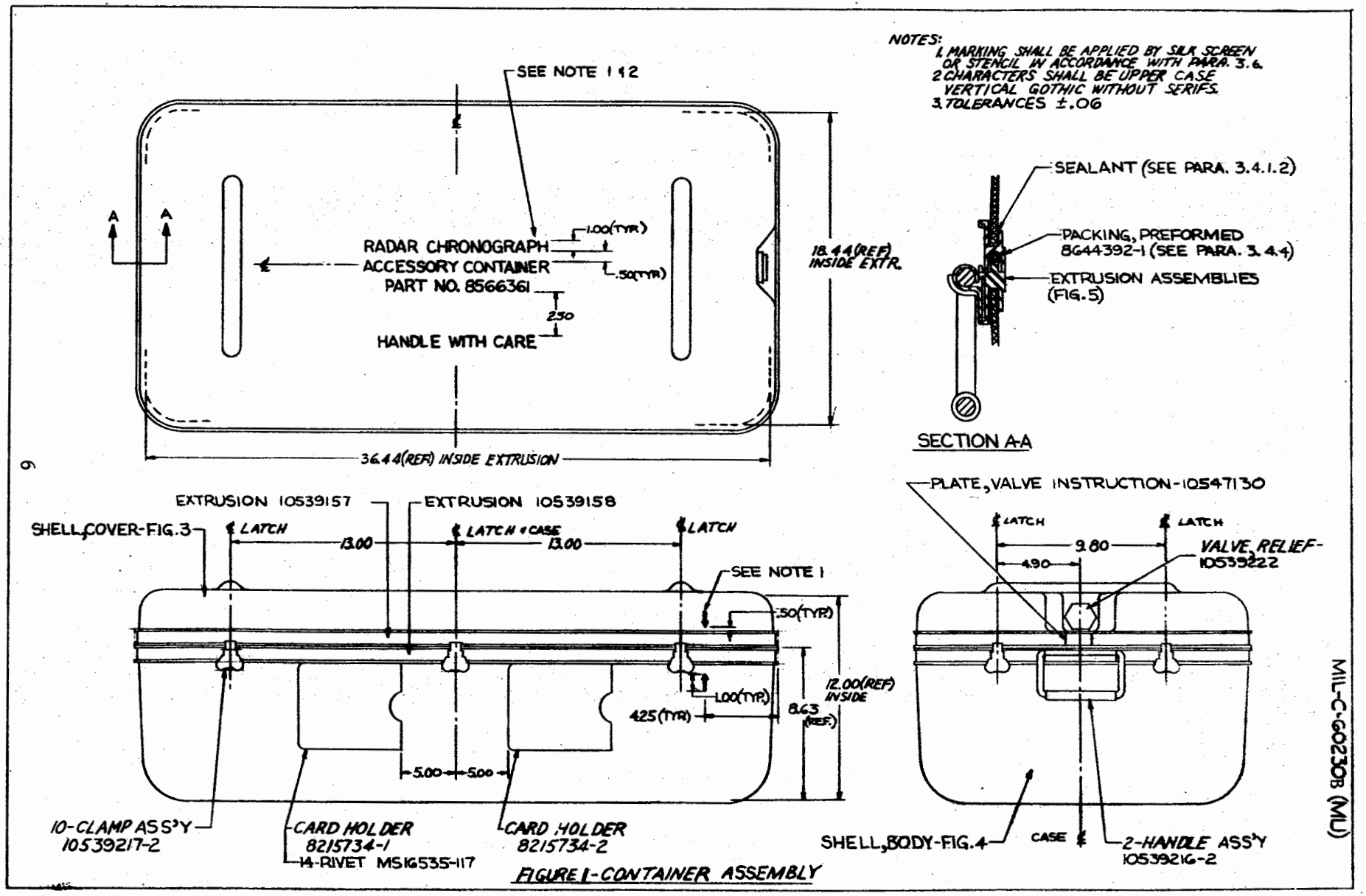


FIGURE 1-CONTAINER ASSEMBLY

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NOTE:
1. CUSHION INSERTS SHALL BE BONDED IN POSITION AS SHOWN WITH ADHESIVE CONFORMING TO MIL-A-5092, TYPE II

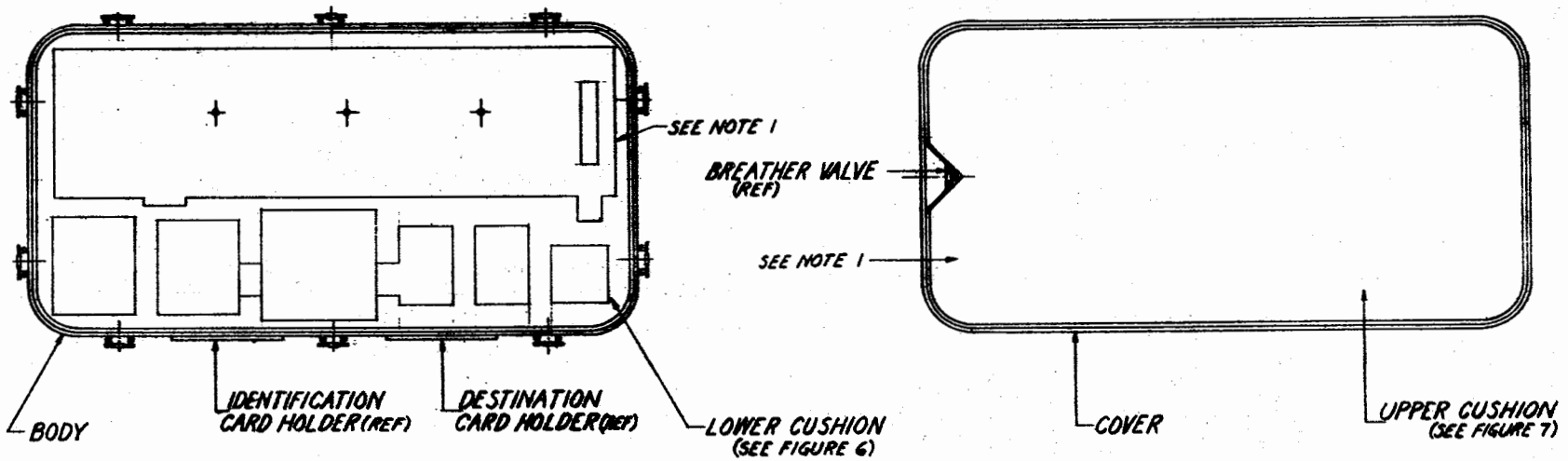
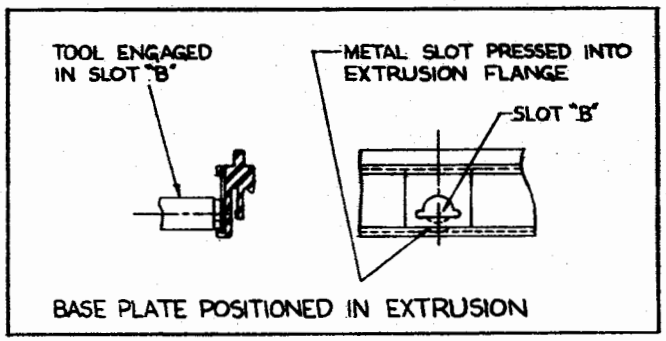


FIG. 1A-CUSHION ARRANGEMENT

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INSTALLATION PROCEDURE:

1. LOCATE CLAMP, CLASP, HANDLE, & HINGE ASSEMBLIES, AS REQUIRED, IN POSITION AS SHOWN IN FIGURE 1.
2. POSITION HARDWARE BASE PLATE IN EXTRUSION FLANGE.
3. ENGAGE INSTALLATION TOOL IN SLOT "B", AS SHOWN.
4. FIRMLY PRESS TOOL DOWNWARD & TURN TOOL 90° TO THE RIGHT & 90° TO THE LEFT, TO DEFORM SLOT OF BASE PLATE IN EXTRUSION FLANGE AS SHOWN.



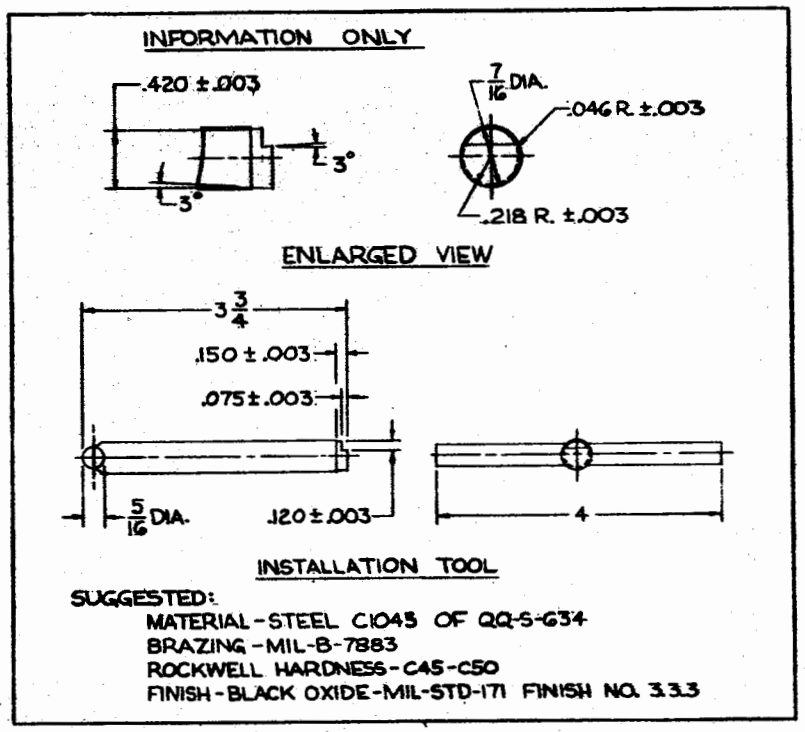
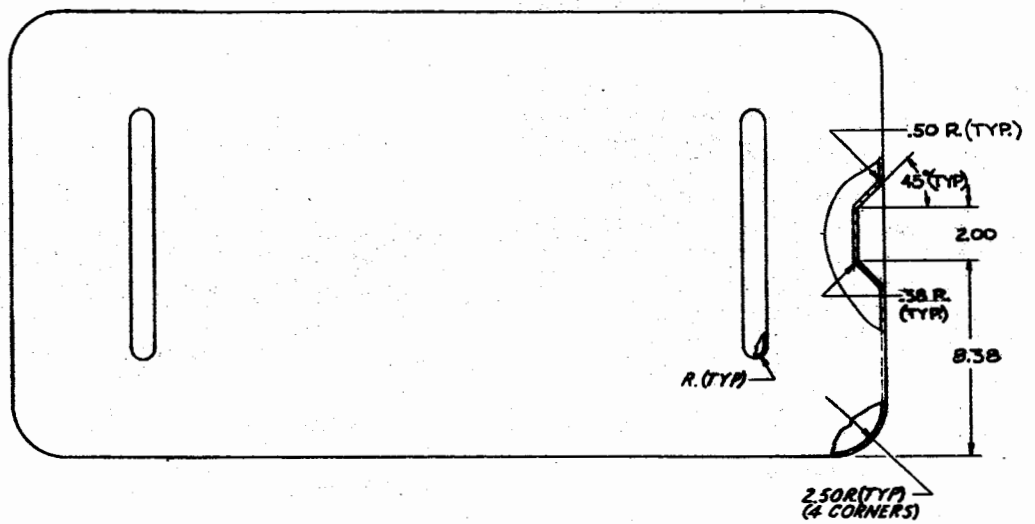


FIGURE 2 - INSTALLATION OF HARDWARE

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NOTES:
1. ALL PRODUCTION SHELLS SHALL BE INTERCHANGEABLE WITHIN .06 AND FIT FIG. 5.
2. MATERIAL SHALL BE AS SPECIFIED IN PARA. 3.2:

   COLOR-OLIVE DRAB-X24087 OF FED-STD-595; FINISH-HAIRCELL GRAIN EXTERIOR, SMOOTH INTERIOR; FORMED WALL THICKNESS .120 +.020,-.030
3. TOLERANCE UNLESS OTHERWISE SPECIFIED: DECIMALS ±.03
   ANGLES ±1°



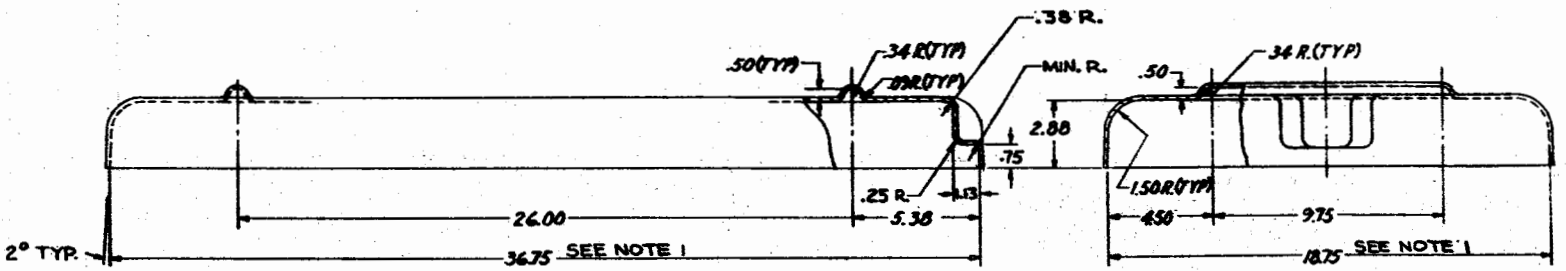


FIGURE 3- SHELL, COVER

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NOTES:
1. ALL PRODUCTION SHELLS SHALL BE INTERCHANGEABLE WITHIN .06 AND FIT FIG. 5.
2. MATERIAL SHALL BE AS SPECIFIED IN PARA. 3.2; COLOR-OLIVE DRAB-X24087 OF FED-STD-595; FINISH-HAIRCELL GRAIN EXTERIOR AND SMOOTH INTERIOR; FORMED WALL THICKNESS .120 +.020, -.030
3. TOLERANCE UNLESS OTHERWISE SPECIFIED: DECIMALS ±.03, ANGLES ±1°

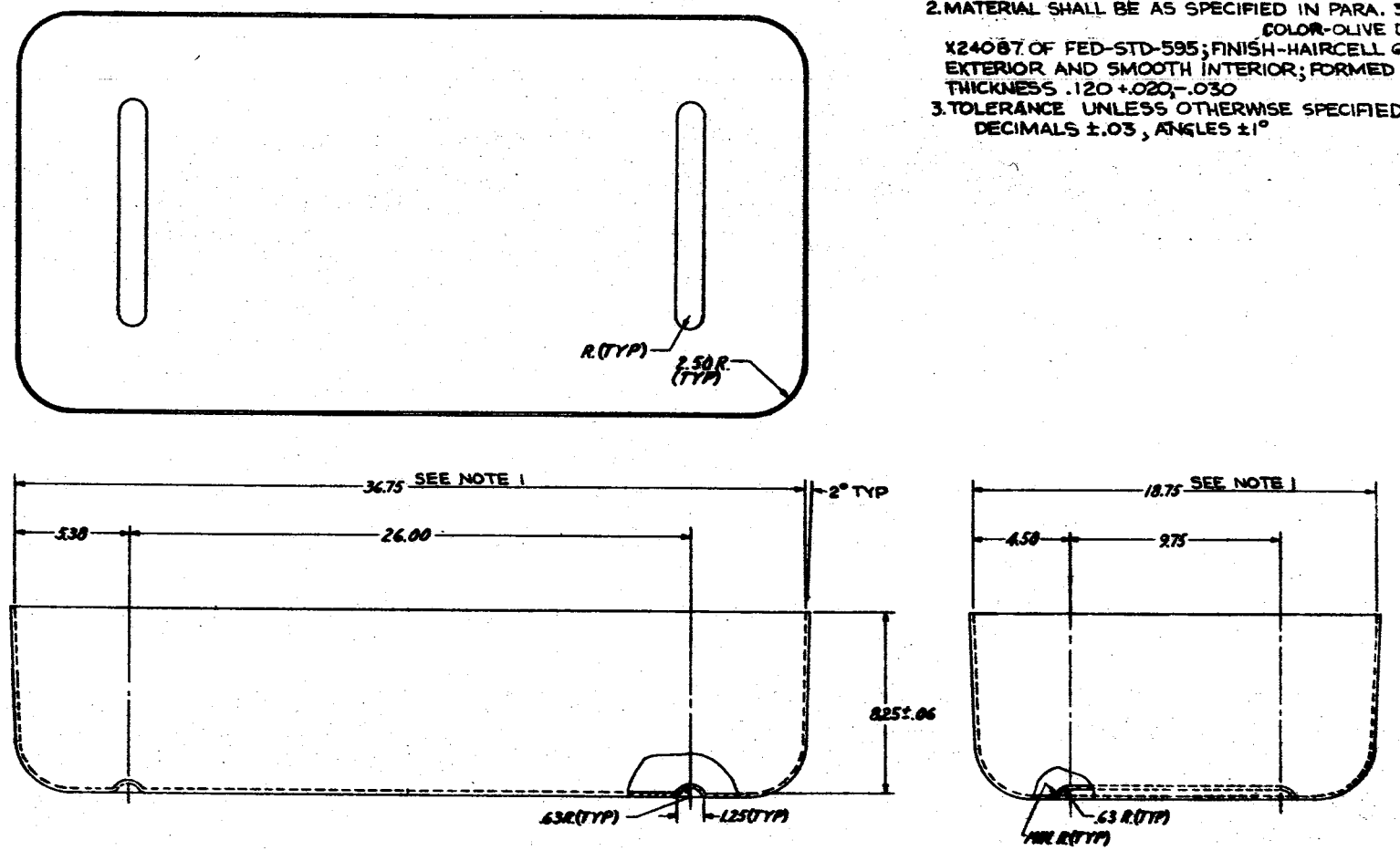
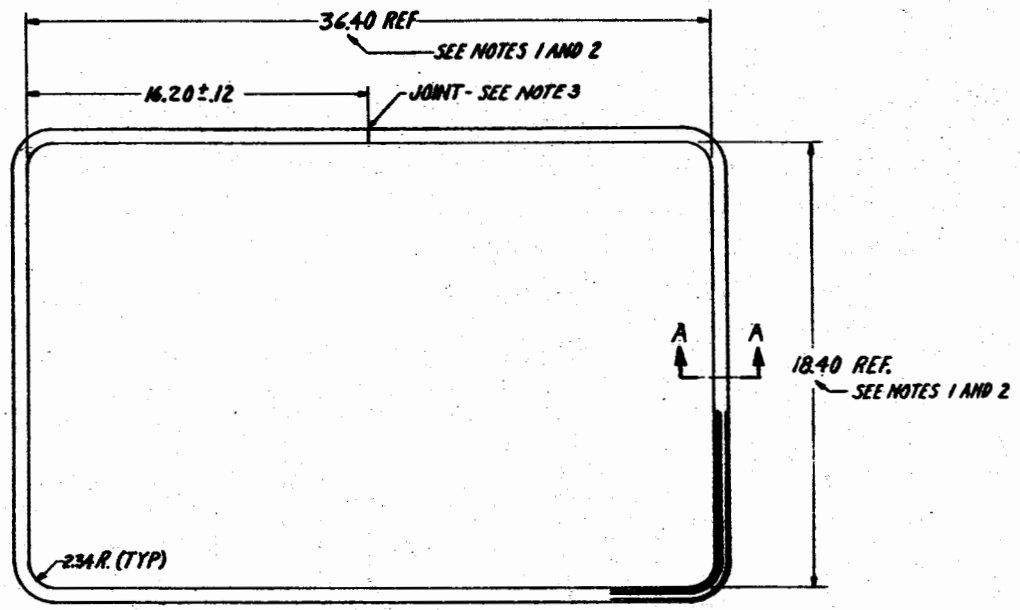


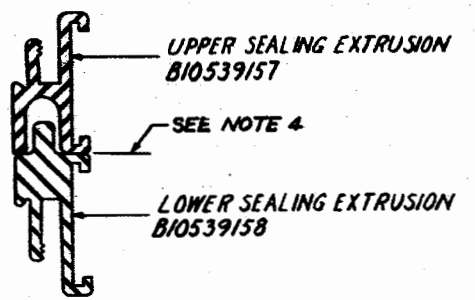
FIGURE 4-SHELL, BODY

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NOTES:

1. COMPLETED EXTRUSIONS SHALL FIT FREELY INTO EACH OTHER AS SHOWN IN SECTION A-A.
2. MUST MATCH SHELLS SHOWN ON FIGURES 3 AND 4.
3. EDGE FRAMES SHALL BE FORMED, WELDED, AND FINISHED IN ACCORDANCE WITH PAR. 3.4.1 AND 3.4.1.1.
4. SURFACE SHOWN SHALL BE FLAT WITHIN .020 INCH T.I.R IN ANY 6 INCHES OF LENGTH AND .030 INCH T.I.R. OVER ENTIRE LENGTH OF ANY SIDE OR END.





SECTION A-A

FIGURE 5- EXTRUSION ASSEMBLIES

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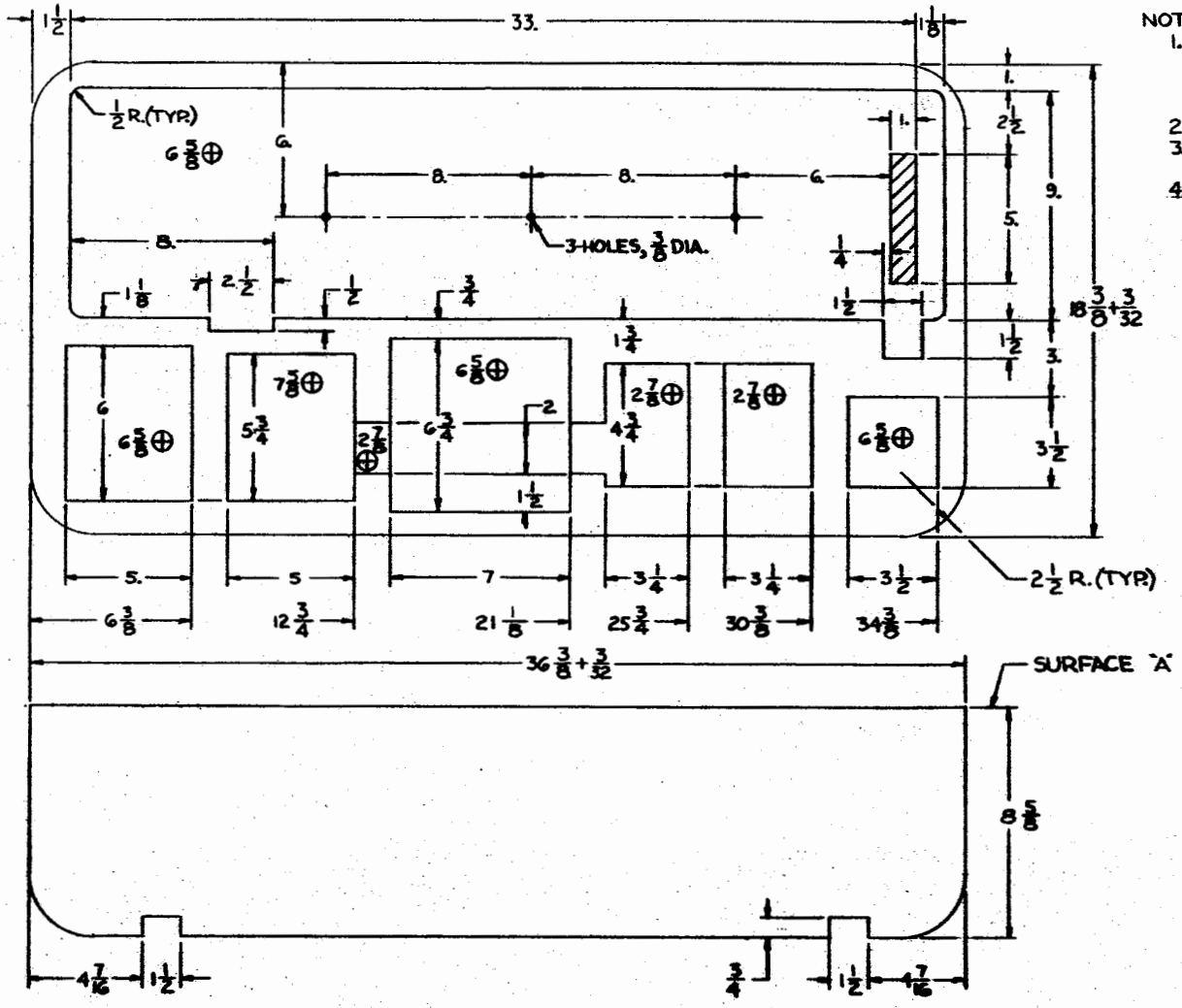


FIG. 6 - LOWER CUSHION

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NOTES:
1. FABRICATE OF POLYETHYLENE FOAM-CONFORMING TO MIL-C-26655-2A, 2 POUNDS/CUBIC FOOT DENSITY, COLOR-CHARCOAL GRAY OR BLACK

2. TOLERANCES: UNLESS OTHERWISE SPECIFIED-
DECIMALS ± .06
ANGLES ± 1°

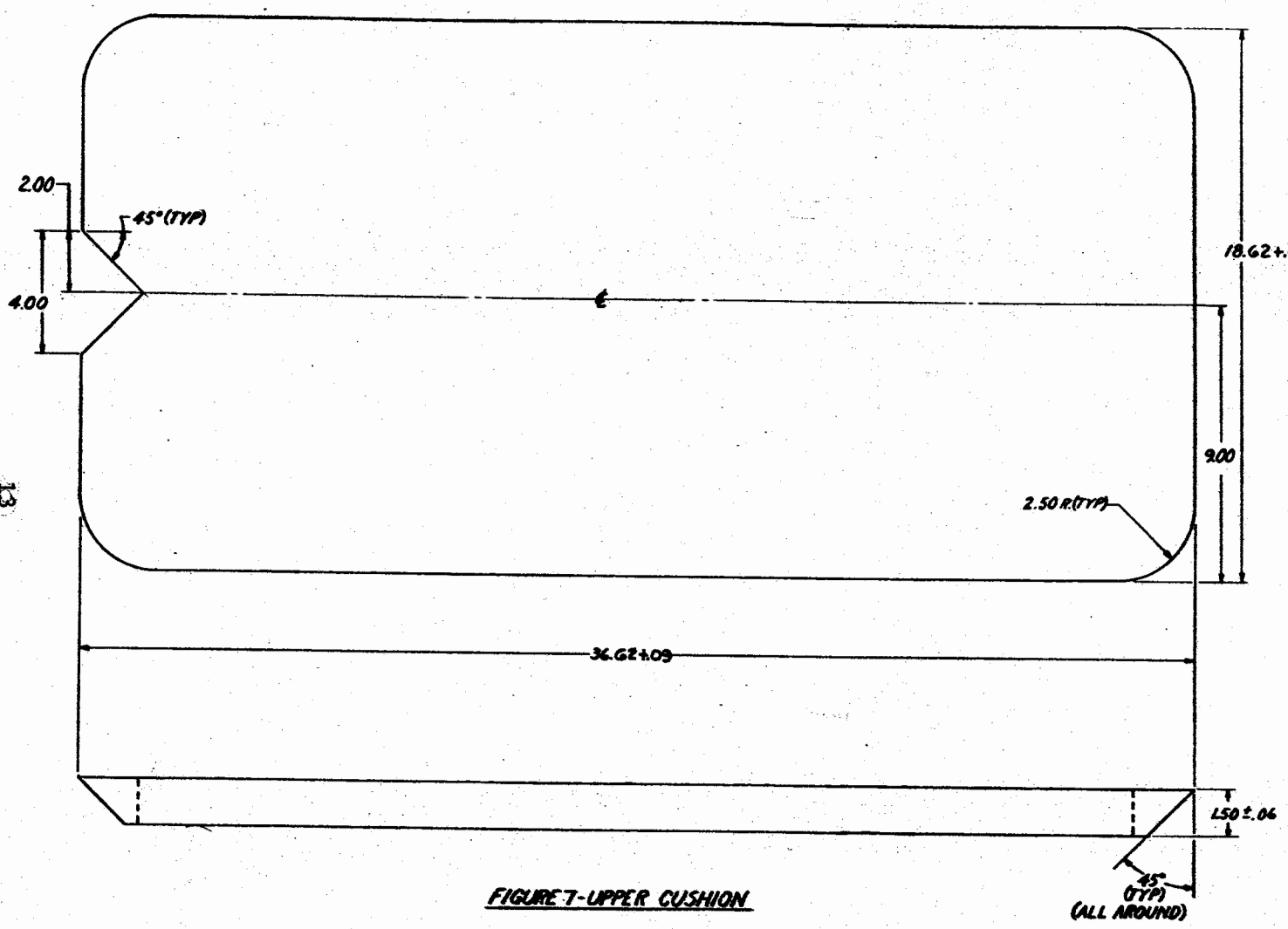


FIGURE 7-UPPER CUSHION

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (Initial Production) Approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of two (2) containers manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Table II. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.0% defective		
101. Improper forming of shells	3.3	Visual, SME
102. Sealing extrusions improperly formed, welded, or brazed	3.4.1	Visual
103. Extrusion assemblies improperly finished	3.4.1.1	Visual
104. Improper sealant, or attachment of extrusions	3.4.1.2	Visual Certification
105. Improper clamps or assembly to container	3.4.2	Visual
106. Improper handles or assembly to container	3.4.3	Visual
107. Improper packing or improperly bonded to extrusion	3.4.4	Visual
108. Improper relief valve improperly installed, instruction plate missing or improperly installed	3.4.5, 3.4.5.1	Visual
109. Card holder missing or improperly installed	3.4.6	Visual
110. Incorrect marking	3.6	Visual
111. Cushion inserts improperly fabricated	3.5.1.1	Visual SME
112. Cushion inserts improperly bonded within container	3.5.1	Visual
MINOR: AQL 4.0% defective		
201. Improper finish	3.2.2	Visual
202. Excess adhesive on packing	3.4.4	Visual

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective

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units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

NO.	CHARACTERISTIC	TEST DATA TO COMPLY WITH
301.	Material	3.2

4.4 Special sampling.

4.4.1 General.- Two container assemblies shall be selected at random as a special sampling from each lot produced. The samples shall have successfully met all other acceptance test specified herein, and shall then meet the following requirements and tests:

TEST		REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1 - 3.6	SME Visual, Figures 1, 1A, 2, 3, 4, 5, 6 and 7
302.	Workmanship	3.9	Visual

4.4.2 Environmental and functional.- Two sample container assemblies shall be selected at random as sample from each lot produced. The samples shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

TEST		REQUIREMENT	TEST PROCEDURE
303.	Impact resistance	3.8.2	4.6.4
304.	Pressure tight	3.7	4.6.1
305.	Storage	3.8.1	4.6.3

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection

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equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall make available inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.2.1 Preparation for tests.- The container assembly shall be fitted with dummy loads simulating the weight and mass of associated equipment for the Radar Chronograph Set, M36 (see 6.1.2). All tests shall be performed with relief valve removed.

4.6 Test methods and procedures.

4.6.1 Pressure retention test.- Containers shall have the relief valves removed and replaced by an appropriate plug. A suitable connection shall be made for the introduction of air pressure. A gage shall be connected to indicate pressure within the container. Air pressure shall be applied up to 1 pound per square inch and the intake valve closed to retain pressure. A soap solution shall be applied to all joints, seams and welds. There shall be no evidence of leakage. After five minutes the gage shall be observed to make sure that the original pressure has been maintained (see 3.7).

4.6.2 Storage.- The container assembly with simulated loads shall be tested as follows:

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(a) Elevated temperature.- The container assembly shall be placed in test chamber and the temperature elevated gradually to 160°F and allowed to remain at this temperature until thermo-stabilization is reached within the container. The test sample shall remain at the thermostabilized temperature of 160°F for a period of 4 hours, after which the temperature shall be gradually reduced to ambient. There shall be no evidence of cracks, or distortion as a result of the test. There shall be no leakage when the container is tested in accordance with 4.6.1.

(b) Low temperature.- The container assembly shall be placed in test chamber and the temperature lowered gradually to -70°F and allowed to remain at this temperature until thermostabilization is reached within the container. The test sample shall remain at the thermostabilized temperature of -70°F for a period of 4 hours, after which the temperature shall be gradually raised to ambient. There shall be no evidence of cracks, delamination or distortion as a result of the test. There shall be no leakage when the container is tested in accordance with 4.6.1.

4.6.3 Impact resistance (ambient temperatures).- The sample container shall be supported at one end of its base on a sill or block 5 inches in height. The opposite end of the container shall be allowed to fall freely from a height of 24 inches. Following the drop test, the container assembly shall be subjected to the pressure retention test specified in 4.6.1.

4.6.4 Impact resistance (low temperature).- The sample container shall be placed in a test chamber and the temperature lowered gradually to -40°F and allowed to remain at this temperature until thermostabilization is reached within the container. The test sample shall remain at the thermostabilized temperature of -40°F for a period of 4 hours. The container shall be immediately subjected to drop test described in 4.6.3. The container shall be subjected to pressure retention test specified in 4.6.1, after container has been stabilized at ambient temperatures.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Unless otherwise specified, the container shall be prepared for shipment using normal commercial methods, or materials to assure acceptance by common or other carrier for safe transportation at the lowest rate to point of destination.

5.2 Marking.- Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use.- The container assembly defined herein is intended for all types of shipment and storage. The container constitutes part of "Set Packaging for Radar Chronograph Set, M36 - Box 2 of 4" as specified in Packaging Data Sheet 10525235.

6.1.1 The container will accommodate the following accessories used in conjunction with the Radar Chronograph:

(1) Tripod, Assembly - 8213856
(1) Receiver, Radar Test - 10525559 w/1 cable 8566401
(1) Microphone - 10531933
(1) Telescope, XM128 - 10540459
  (1) Case, Carrying, M87 - 7654351
(1) Bracket, Lamp Assembly - 7641600
(1) Kit, Repair Parts - 5800940
(1) Equipment Log Book Binder
(1) TM Manual (TM 9-1290-325-12/1)
(1) TM Manual (TM 9-1290-325-12/2)
(1) Chronograph Automatic Reliability Rater - 10552370
(1) Light Instrument M53E1 - 10553463

6.1.2 Dummy loads.- Dummy loads referenced in 4.5.2.2.1 shall have the following weight and mass:

MASS	WEIGHT
(1) 34 by 9 by 8-1/2	25 pounds
(1) 6 by 5 by 8-1/2	5 pounds
(1) 5-3/4 by 5 by 9-1/2	5 pounds
(1) 7 by 6-3/4 by 8-1/2	5 pounds
(1) 3-1/2 by 3-1/2 by 8-1/2	3 pounds

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Applicable stock number
(c) Requirement for submission of material swatch (3.2.1)

6.3 Federal stock number.- FSN 1290-804-9421.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1290-A315

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